# JCEI'S JAIHIND COLLEGE OF ENGINEERING, KURAN

## A.Y. 2025-26

MINI PROJECT ON

**"ONLINE VOTING SYSTEM''**

UNDER

**"DATABASE MANAGEMENT SYSTEM"**

SE-AI&DS ENGINEERING

DEPARTMENT OF AI & DS ENGINEERING

SEMESTER-IV

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# Department of Artificial Intelligence and data science Engineering

**(2025-26)**

# Certificate

This is to certify that,

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I have successfully completed the Mini project entitled **"Online Voting System"** Under the guidance of **Prof.P.K.Karpe** in partial fulfillment of the requirements for the Second Year of Engineering in Artificial intelligence and Data Science under the Savitribai Phule Pune University during the academic year 2025-2026.

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Yours faithfully,

**Ms.Shete Priti Rahul**

# ABSTRACT

The Online Voting System is a modern digital application designed to conduct elections in a secure, efficient, and transparent manner. Unlike traditional paper-based voting, which is time-consuming, costly, and prone to human errors, this system automates the entire process from voter authentication to result declaration. Registered voters can cast their votes from any location using an internet-enabled device, making the system highly convenient and accessible. Each voter is assigned unique login credentials to ensure secure authentication and to prevent unauthorized access or multiple voting. The system uses a centralized database, such as MySQL, to store voter details, candidate information, and votes, ensuring data integrity and consistency through DBMS concepts like primary and foreign keys. It also includes basic security measures to protect sensitive data. One of its key features is automatic vote counting, which eliminates manual errors and provides instant, accurate results. Additionally, it reduces operational costs and paper usage, making it eco-friendly. With a user-friendly interface, the system is easy to use and can be further enhanced with technologies like encryption and biometric authentication.

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# 1. INTRODUCTION

Voting is a fundamental process used in decision-making across various domains such as governments, educational institutions, and organizations. It allows individuals to express their opinions and select representatives or make collective decisions. Traditionally, voting is conducted using paper ballots and manual counting, which often leads to several challenges such as time consumption, high operational costs, human errors, and lack of transparency.

The Online Voting System is developed to overcome these limitations by providing a digital and automated solution for conducting elections. It is a software-based system that allows voters to cast their votes securely through an online platform using computers or mobile devices. This system eliminates the need for physical presence at polling stations and simplifies the entire voting process.

In the traditional system, maintaining voter records, verifying identities, and counting votes require significant manpower and resources. There is also a risk of invalid votes, duplication, and manipulation. The Online Voting System addresses these issues by implementing secure authentication mechanisms, ensuring that only authorized users can access the system and vote. Each voter is assigned unique login credentials, and the system ensures that a voter can cast only one vote.

The system uses a centralized database to store all relevant information, including voter details, candidate information, and voting records. By applying Database Management System (DBMS) concepts such as primary keys, foreign keys, and normalization, the system maintains data integrity, consistency, and security. The use of SQL queries allows efficient storage, retrieval, and management of data.

One of the major advantages of this system is the automation of vote counting. Unlike manual counting, which is time-consuming and error-prone, the system calculates results instantly and accurately. This enhances transparency and reduces the chances of disputes. Additionally, the system can generate reports and display results in a clear and organized manner.

The Online Voting System is designed with a user-friendly interface, making it easy to use even for individuals with basic technical knowledge. It is also cost-effective, as it reduces the need for paper, transportation, and manpower. Moreover, it contributes to environmental sustainability by minimizing paper usage.

This project demonstrates the practical application of programming and database concepts in solving real-world problems. It provides a secure, efficient, and scalable solution for conducting elections and can be further enhanced with advanced features such as encryption, biometric authentication, and web or mobile-based access for larger-scale implementation.

# 2. OBJECTIVES

The main objective of the Online Voting System is to develop a secure, reliable, and efficient platform for conducting elections digitally. The specific objectives of the system are as follows:

**1. To Develop a Secure Voting System**

To ensure that only authorized voters can access the system through proper authentication mechanisms such as login credentials.

**2. To Eliminate Manual Voting Process**

To replace traditional paper-based voting with an automated system, reducing paperwork and human effort.

**3. To Prevent Duplicate Voting**

To ensure that each voter can cast only one vote by implementing validation and database constraints.

**4. To Automate Vote Counting**

To provide instant and accurate results without the need for manual counting.

**5. To Maintain Data Accuracy and Integrity**

To store and manage voting data using DBMS concepts like primary keys and foreign keys, ensuring consistency and reliability.

**6. To Provide Centralized Data Management**

To store all voter, candidate, and voting records in a single database for easy access and management.

**7. To Improve Transparency in Elections**

To ensure fairness and build trust by providing clear and accurate results.

**8. To Reduce Cost and Time**

To minimize expenses related to paper, manpower, and logistics, while speeding up the voting process.

# 3. REQUIREMENT SPECIFICATION

The requirement specification defines the necessary hardware and software components required for the proper functioning of the Online Voting System. It ensures that the system runs efficiently and meets all functional and performance needs.

**1. Hardware Requirements**
The system does not require high-end hardware and can run on standard computing devices. The basic hardware requirements are:
A computer or laptop with a minimum Intel i3 processor or equivalent
At least 4 GB RAM for smooth performance
Minimum 10 GB free disk space
Stable internet connection for online access
Input devices such as keyboard and mouse

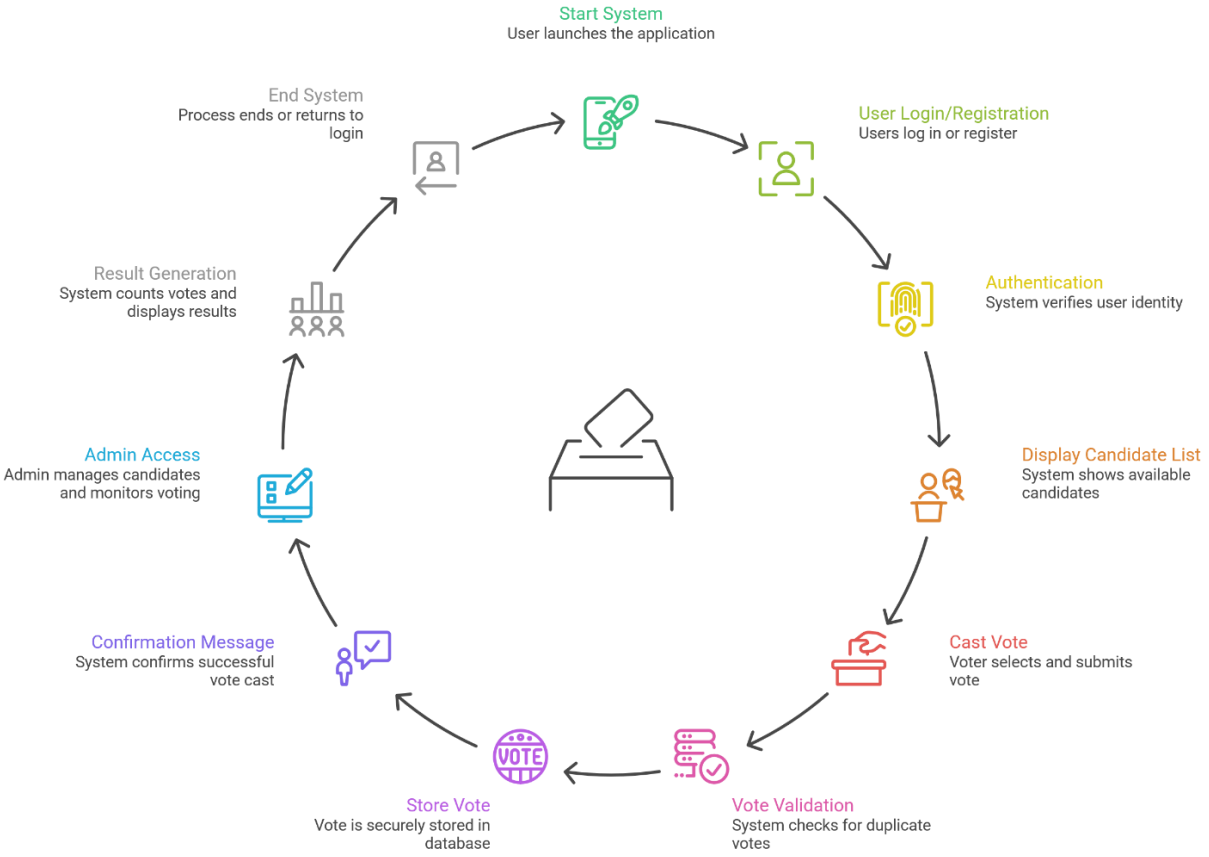
**2. Software Requirements**
The software components required for developing and running the system include:
a) Programming Language
Python / Java
These languages are used to develop the application logic. Python is preferred for its simplicity and readability, while Java provides robustness and security.
b) Database
MySQL
MySQL is a relational database management system used to store voter details, candidate information, and voting records. It ensures data consistency, security, and efficient data retrieval.
c) Interface
Command Line Interface (CLI) or Web-Based Interface
The system can be implemented using a simple CLI for basic operations or upgraded to a web interface for better user experience and accessibility.
d) Libraries and Tools
MySQL Connector (to connect application with database)
Flask / JSP (optional) for web-based implementation
HTML, CSS (optional) for front-end design

**3. Functional Requirements**
Voter registration and login system
Candidate management by admin
Secure voting process
Prevention of duplicate voting
Automatic vote counting
Result display and reporting

**4. Non-Functional Requirements**
Security: Protect voter data and prevent unauthorized access
Performance: Fast response and quick result generation
Reliability: Accurate and consistent system behavior
Usability: Easy-to-use interface
Scalability: Ability to handle increasing number of users

# 4. WORKFLOW

**1. Start System**
The user launches the application or opens the voting platform.
**2. User Login / Registration**
New users register by entering details.
Existing users log in using their credentials (ID & Password).
**3. Authentication**
The system verifies the voter's identity to ensure only authorized users can access the voting system.
**4. Display Candidate List**
After successful login, the system shows the list of candidates available for voting.
**5. Cast Vote**
The voter selects a candidate and submits their vote.
**6. Vote Validation**
The system checks whether the voter has already voted to prevent duplicate voting.
**7. Store Vote in Database**
If valid, the vote is securely stored in the database.
**8. Confirmation Message**
The system displays a message like "Vote Cast Successfully".
**9. Admin Access (Optional)**
Admin can view/manage candidates and monitor voting.
**10. Result Generation**
After voting ends, the system counts votes automatically and displays results.
**11. End System**
The process ends or returns to login for the next user.



# 5. FUNCTIONALITY

The Online Voting System provides various features that ensure smooth, secure, and efficient election management. The major functionalities of the system are as follows:

**1. Voter Registration**

- New users can register themselves in the system.
- The system collects details such as voter name, ID, and password.
- Each voter is assigned a unique Voter ID.
- The data is securely stored in the database.

**2. User Login (Authentication)**

- Registered voters can log in using their credentials.
- The system verifies the username and password.
- Only authenticated users are allowed to access the voting system.

**3. Candidate Management**

- The admin can add, update, or delete candidate details.
- Candidate information includes name, ID, and party.
- This ensures that the candidate list is always up-to-date.

**4. Display Candidate List**

- After successful login, voters can view the list of candidates.
- The list is displayed in a clear and organized format.
- Helps voters make informed decisions.

**5. Vote Casting**

- Voters can select their preferred candidate and cast their vote.
- The system records the vote securely.
- Simple interface ensures ease of use.

**6. Vote Validation**

- The system checks whether the voter has already voted.
- Prevents duplicate or multiple voting.
- Ensures fairness and integrity of the election.

**7. Vote Storage**

- All votes are stored in a centralized database.
- Uses DBMS concepts like primary and foreign keys.
- Ensures data consistency and security.

**8. Result Generation**

- The system automatically counts votes.
- Results are generated instantly after voting ends.
- Displays total votes received by each candidate.

**9. Admin Control Panel**

- Admin can manage voters and candidates.
- Admin can monitor voting process and view results.
- Provides full control over the system.

**10. Exit System**

- Allows users to safely log out or exit the application.
- Ensures proper termination of the system session.

# 6. CONCEPTUAL DESIGN

The conceptual design represents the overall structure of the Online Voting System. It identifies themain entities, their attributes, and the relationships between them without focusing on implementation details. This design helps in understanding how data flows and interacts within the system.

**6.1 Entities and Attributes**

The system consists of the following main entities:

1. Voter
   - Voter_ID (Primary Key)
   - Name
   - Password
   - Status (Voted/Not Voted)

   This entity stores details of all registered voters.
2. Candidate
   - Candidate_ID (Primary Key)
   - Candidate_Name
   - Party

   This entity contains information about all candidates participating in the election.
3. Vote
   - Vote_ID (Primary Key)
   - Voter_ID (Foreign Key)
   - Candidate_ID (Foreign Key)

   This entity records each vote cast by a voter.
4. Admin (Optional)
   - Admin_ID (Primary Key)
   - Username
   - Password

   This entity manages system operations like adding candidates and viewing results.

**6.2 Relationships**

The relationships between entities are defined as follows:

Voter → Vote
- One voter can cast only one vote.
- Relationship: One-to-One

Candidate → Vote
- One candidate can receive many votes.
- Relationship: One-to-Many

Admin → Candidate
- Admin manages multiple candidates.
- Relationship: One-to-Many

**6.3 ER Diagram Explanation**

- The Voter entity is connected to the Vote entity through Voter_ID.
- The Candidate entity is connected to the Vote entity through Candidate_ID.
- The Vote entity acts as a bridge between Voter and Candidate.
- Primary keys uniquely identify each record, while foreign keys maintain relationships.

**6.4 Design Approach**

- Data is organized into separate tables to avoid redundancy.
- Normalization techniques are used to maintain data consistency.
- Relationships are maintained using primary and foreign keys.

**ER Features**

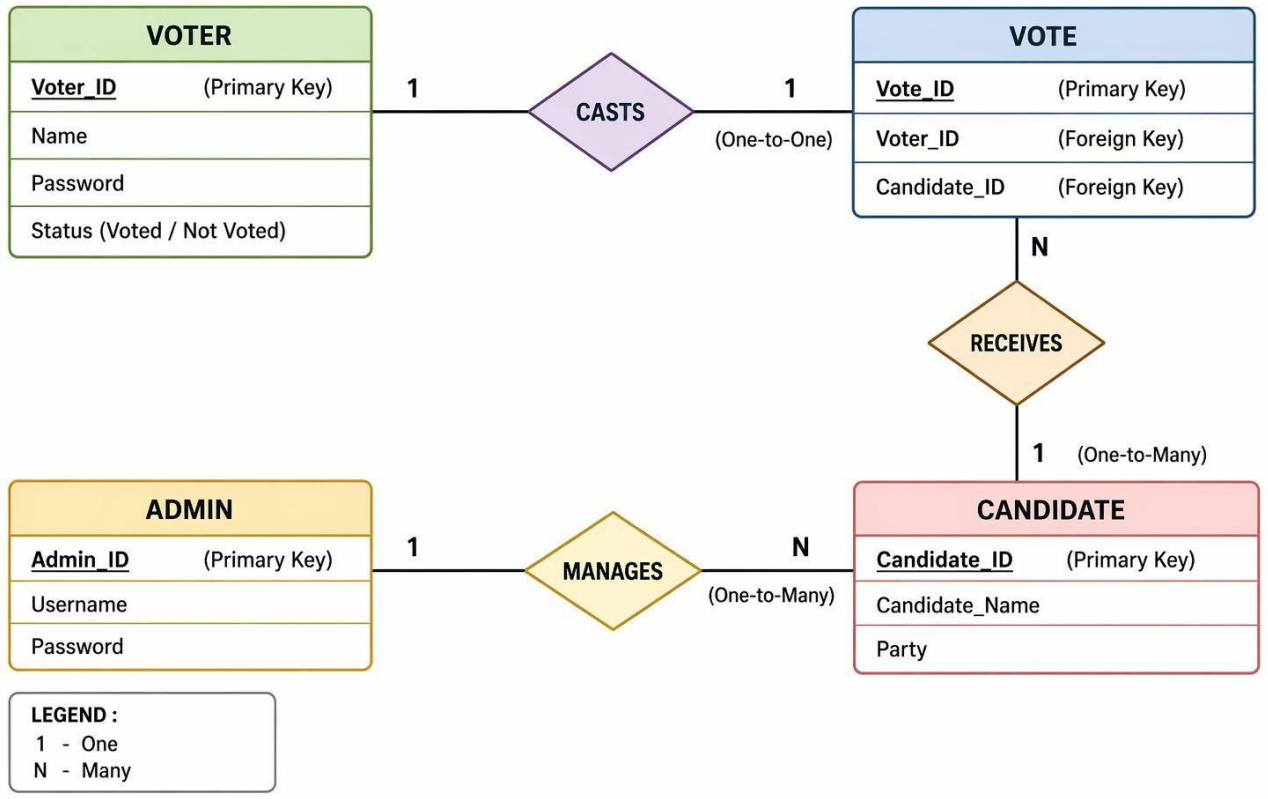



**Fig. ER Diagram for Online Voting System**

# 7. IMPLEMENTATION

## 1. Python Code

```
!pip install mysql-connector-python

import mysql.connector

db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="12345678",  # your MySQL password
    database="election_system"
)

cursor = db.cursor()
print("Connected to MySQL")
n = int(input("Enter number of voters: "))

voter_ids = []

for i in range(n):
    print(f"\nVoter {i+1}")
    name = input("Enter Name: ")
    email = input("Enter Email: ")
    password = input("Enter Password: ")

    cursor.execute(
        "INSERT INTO voter_info(name,email,password) VALUES (%s,%s,%s)",
        (name, email, password)
    )
    db.commit()

    voter_ids.append(cursor.lastrowid)
    print("Voter Registered")

print("\nAll voters registered!\n")
cursor.execute("SELECT * FROM candidate_info")
candidates = cursor.fetchall()

print(" Candidates:")
for c in candidates:
    print(f"{c[0]} : {c[1]}")
for vid in voter_ids:
    print(f"\nVoter ID {vid} voting:")

    choice = int(input("Enter Candidate ID: "))

    # Check if already voted
    cursor.execute("SELECT * FROM vote_records WHERE voter_id=%s", (vid,))
    if cursor.fetchone():
        print(" Already voted!")
```

```
    else:
        cursor.execute(
            "INSERT INTO vote_records(voter_id,candidate_id) VALUES (%s,%s)",
            (vid, choice)
        )
        db.commit()

        print("Vote Cast")
cursor.execute("""
SELECT c.candidate_name, COUNT(v.vote_id)
FROM candidate_info c
LEFT JOIN vote_records v ON c.candidate_id = v.candidate_id
GROUP BY c.candidate_id
""")

results = cursor.fetchall()

print("\n Election Results:")
for r in results:
    print(f"{r[0]} : {r[1]} votes")

db.close()
```

## 2. MySQL

```
CREATE DATABASE election_system;
USE election_system;

-- Voter Table
CREATE TABLE voter_info (
    voter_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50),
    password VARCHAR(50)
);

-- Candidate Table
CREATE TABLE candidate_info (
    candidate_id INT AUTO_INCREMENT PRIMARY KEY,
    candidate_name VARCHAR(50)
);

-- Vote Table
CREATE TABLE vote_records (
    vote_id INT AUTO_INCREMENT PRIMARY KEY,
    voter_id INT,
    candidate_id INT,
    FOREIGN KEY (voter_id) REFERENCES voter_info(voter_id),
    FOREIGN KEY (candidate_id) REFERENCES candidate_info(candidate_id)
);
```

```
-- Insert Candidates
INSERT INTO candidate_info(candidate_name)
VALUES ('Ram'), ('Sham'), ('Tukaram');
```

```
SELECT * FROM voter_info;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| voter_id | name | email | password |
|---|---|---|---|
| 1 | Shete Priti | sheteprit@gmail.com | 1234 |
| 2 | Sherkar Manisha | sherkarmanisha@gmail.com | 456 |
| 3 | Raut Riya | rautriya@gmail.com | 789 |
| 4 | priti | sheteprit@gmail.com | 123 |
| 5 | manisha | sherkarmanisha@gmail.com | 456 |
| 6 | riya | rautriya@gmail.com | 756 |
| 7 | manali | nali@gmail.com | 547 |
| NULL | NULL | NULL | NULL |

voter_info 8 | Apply | Revert

```
SELECT * FROM vote_records;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| vote_id | voter_id | candidate_id |
|---|---|---|
| 1 | 1 | 2 |
| 2 | 2 | 3 |
| 3 | 3 | 2 |
| 4 | 4 | 3 |
| 5 | 5 | 1 |
| 6 | 6 | 2 |
| 7 | 7 | 3 |
| NULL | NULL | NULL |

vote_records 9 | Apply | Revert

```
SELECT c.candidate_name, COUNT(v.vote_id)
FROM candidate_info c
LEFT JOIN vote_records v ON c.candidate_id = v.candidate_id
GROUP BY c.candidate_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| candidate_name | COUNT(v.vote_id) |
|---|---|
| Ram | 1 |
| Sham | 3 |
| Tukaram | 3 |

Result 10 | Read Only

# 8. TESTING

The testing phase confirmed that the system is reliable, secure, and efficient. It performs all operations correctly and meets the required objectives. Any minor errors found during testing were corrected, ensuring smooth system performance.



In [1]:
```
!pip install mysql-connector-python
```

```
Requirement already satisfied: mysql-connector-python in c:\users\admin\anaconda3\li
b\site-packages (9.6.0)
```

In [2]:
```
import mysql.connector

db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="12345678",
    database="election_system"
)

cursor = db.cursor()
print("Connected to MySQL")
```

```
Connected to MySQL
```

In [3]:
```
n = int(input("Enter number of voters: "))

voter_ids = []

for i in range(n):
    print(f"\nVoter {i+1}")
    name = input("Enter Name: ")
    email = input("Enter Email: ")
    password = input("Enter Password: ")

    cursor.execute(
        "INSERT INTO voter_info(name,email,password) VALUES (%s,%s,%s)",
        (name, email, password)
    )
    db.commit()

    voter_ids.append(cursor.lastrowid)
    print(" Voter Registered")

print("\n All voters registered!\n")
```

```
Voter 1
 Voter Registered

Voter 2
 Voter Registered

Voter 3
 Voter Registered

Voter 4
 Voter Registered

 All voters registered!
```

In [4]:
```
cursor.execute("SELECT * FROM candidate_info")
candidates = cursor.fetchall()

print("Candidates:")
for c in candidates:
    print(f"{c[0]} : {c[1]}")
```

```
Candidates:
1 : Ram
2 : Sham
3 : Tukaram
```

In [5]:
```
for vid in voter_ids:
    print(f"\nVoter ID {vid} voting:")

    choice = int(input("Enter Candidate ID: "))

    # Check if already voted
    cursor.execute("SELECT * FROM vote_records WHERE voter_id=%s", (vid,))
    if cursor.fetchone():
        print("Already voted!")
    else:
        cursor.execute(
            "INSERT INTO vote_records(voter_id,candidate_id) VALUES (%s,%s)",
            (vid, choice)
        )
        db.commit()

        print("Vote Cast")
```

```
Voter ID 4 voting:
Vote Cast

Voter ID 5 voting:
Vote Cast

Voter ID 6 voting:
Vote Cast

Voter ID 7 voting:
Vote Cast
```

In [6]:
```
cursor.execute("""
SELECT c.candidate_name, COUNT(v.vote_id)
FROM candidate_info c
LEFT JOIN vote_records v ON c.candidate_id = v.candidate_id
GROUP BY c.candidate_id
""")

results = cursor.fetchall()

print("\n Election Results:")
for r in results:
    print(f"{r[0]} : {r[1]} votes")

db.close()
```

```
 Election Results:
Ram : 1 votes
Sham : 3 votes
Tukaram : 3 votes
```

In [ ]:

# 9. CONCLUSION

The Online Voting System successfully provides a secure, efficient, and reliable solution for conducting elections in a digital environment. It overcomes the limitations of traditional voting methods by eliminating paperwork, reducing manual effort, and minimizing human errors. The system ensures that each voter can cast only one vote through proper authentication and validation mechanisms, thereby maintaining fairness and integrity in the election process.

By using a centralized database, the system efficiently stores and manages voter details, candidate information, and voting records. The implementation of Database Management System (DBMS) concepts such as primary keys, foreign keys, and normalization ensures data consistency, accuracy, and security. Additionally, the automated vote counting feature enables instant result generation, saving time and improving transparency.

The system is designed with a user-friendly interface, making it easy to use for individuals with basic technical knowledge. It is also cost-effective and environmentally friendly, as it reduces the need for paper and other physical resources.

Overall, the Online Voting System demonstrates the practical application of programming and database technologies in solving real-world problems. It enhances the efficiency of the election process and provides a scalable solution that can be further improved with advanced features such as encryption, biometric authentication, and web or mobile-based access.

# 10. FUTURE SCOPE

The Online Voting System can be further enhanced and expanded with advanced features to improve its functionality, security, and usability. Some of the possible future improvements are as follows:

**1. Biometric Authentication**

The system can be integrated with biometric technologies such as fingerprint or facial recognition to enhance security and ensure that only genuine voters can cast their votes.

**2. Web-Based and Mobile Application**

The system can be developed into a full-fledged web or mobile application, allowing users to access the voting platform anytime and anywhere using smartphones or browsers.

**3. Blockchain Technology Integration**

Blockchain can be used to provide a highly secure and tamper-proof voting system. It ensures transparency and prevents manipulation of votes.

**4. End-to-End Encryption**

Advanced encryption techniques can be implemented to secure data transmission and protect sensitive information such as voter credentials and voting records.

**5. Multi-Level Authentication**

Adding OTP (One-Time Password) or email verification can improve security by providing an additional layer of authentication.

**6. Real-Time Result Analytics**

The system can include dashboards with charts and graphs to display voting statistics and trends in real time.

**7. Multi-Language Support**

The system can support multiple languages to make it accessible to users from different regions and backgrounds.

**8. Cloud-Based Deployment**

Hosting the system on cloud platforms can improve scalability, reliability, and accessibility while supporting a large number of users simultaneously.

**9. Integration with Government ID Systems**

The system can be linked with official identification systems (like Aadhaar) to verify voter identity more effectively.

**10. Improved User Interface (UI/UX)**

Enhancing the design with a graphical user interface (GUI) will make the system more interactive, user-friendly, and visually appealing.

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